SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – CONCEPT MAPPING

Course Code:	CSA6502	**Course Title:**	Generative AI and Large Scale Model
CO Assessed:	CO1-Imitation (BL3)	**Assessment:**	Assessment Tool 1– CONCEPT MAPPING
Weightage:	10% of CIA	**Total Marks:**	25
CO1 Statement:	Explain the fundamentals of Artificial Intelligence, Generative AI, Foundation Models, Transformer architecture, tokenization, embeddings, and Large Language Models.		
Student Name:	A. Mounish	**Reg. No.:**	192472273
Section / Batch:	CSE AI	**Date:**	16/07/2026

Code of Conduct

*I **A. Mounish / 192472273** certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: ___________________________

Instructions

- This assessment contains 6 questions.
- Each student assigned with two questions .
- Each question carries 10 marks. Total: 20 Marks.
- Construct concept maps clearly for each question.
- Show relationships between key concepts using arrows/links.
- Include tokens, rules, transitions, and classifications wherever required.
- Ensure clarity, structure, and logical flow in diagrams.
- Duration: 20 Minutes.

Questions	Concept Identification (25)	Linkages (25)	Organization (20)	Integration of Literature (20)	Clarity (10)	Total Marks
					)verall Total 100)	

Annexure A CONCEPT MAPPING

1.A startup is developing an AI-powered healthcare assistant capable of predicting diseases, generating medical reports, and answering patient queries. Create a concept map showing how Artificial Intelligence, Machine Learning, Deep Learning, Generative AI, and Large Language Models (LLMs) are interconnected in this application. Explain the role of each technology.

AI-POWERED HEALTHCARE ASSISTANT

Artificial Intelligence   Machine Learning   Generative AI

Decision Making   Disease Prediction   Medical Report Generation

► Deep Learning ◄

Medical Image Analysis

Large Language Models

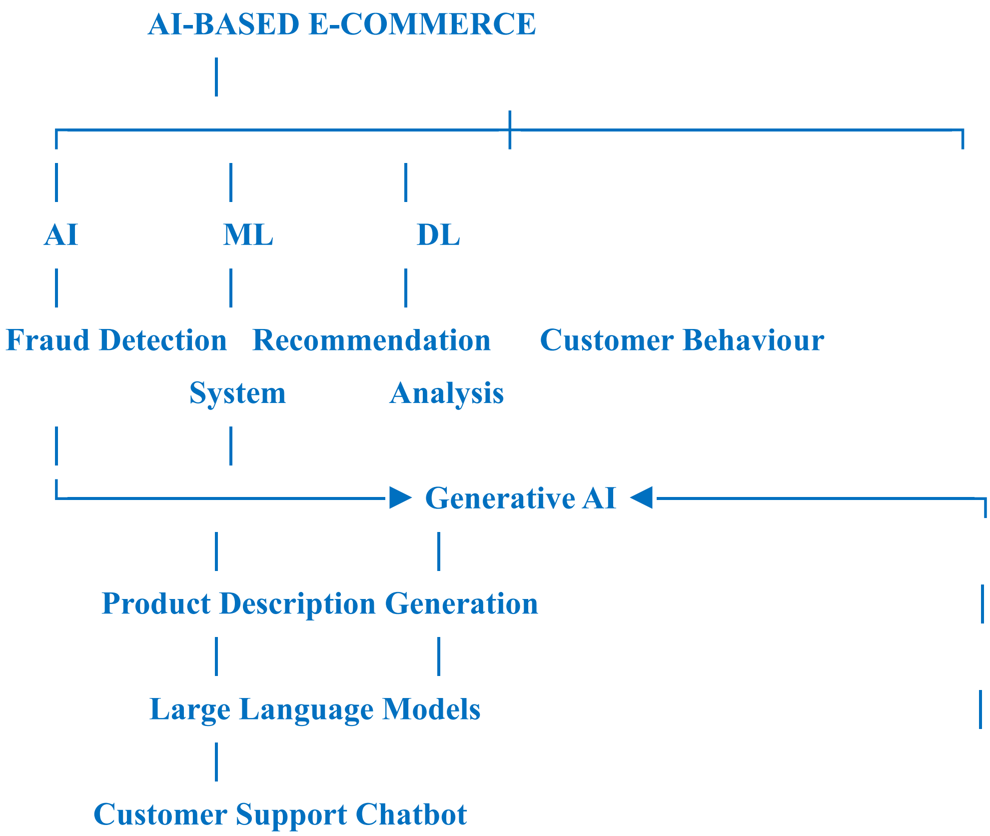
Patient Queries → Medical Advice → Chatbot

Final Healthcare System

Roles:

- **AI – Intelligent decision making**
- **ML – Disease prediction**
- **DL – Image analysis**
- **Generative AI – Report generation**
- **LLM – Patient interaction and question answering**

2. **An e-commerce company uses recommendation systems, chatbot support, fraud detection, and AI-generated product descriptions. Develop a concept map illustrating which components belong to AI, ML, DL, Generative AI, and LLMs. Justify your classification.**



Justification:

- **AI → Decision making**
- **ML → Product recommendation**
- **DL → Pattern recognition**
- **Generative AI → Product description generation**
- **LLM → Chatbot support**

3. **Design a comprehensive mind map for the evolution of intelligent systems, beginning with Artificial Intelligence and progressing through Machine Learning, Deep Learning, Generative**

AI, and Large Language Models. Include key algorithms, applications, and relationships between each technology.

Artificial Intelligence

|

▼

Machine Learning

|

▼

Deep Learning

|

▼

Generative AI

|

▼

Large Language Models

Algorithms

- **AI → Rule-based systems**
- **ML → Decision Tree, Random Forest**
- **DL → CNN, RNN**
- **Generative AI → GAN, Diffusion Models**
- **LLM → Transformer**

Applications

- **AI → Automation**
- **ML → Prediction**
- **DL → Image & Speech Recognition**
- **Generative AI → Content Generation**
- **LLM → Chatbots & Text Generation**

4. **A university plans to introduce an AI-based learning platform featuring personalized recommendations, automated grading, content generation, and an intelligent tutor. Draw a concept map identifying the contribution of AI, ML, DL, Generative AI, and LLMs in the system. Evaluate why all five technologies are necessary instead of using only one.**

AI-BASED LEARNING PLATFORM

AI ML DL Generative AI LLM

Automation Personalized Image Notes & Intelligent
Recommendation Analysis Content Tutor
Generation

Contribution:

- **AI → Automation**
- **ML → Personalized recommendations**
- **DL → Intelligent analysis**
- **Generative AI → Content generation**
- **LLM → Intelligent tutor**

5. **Create a comparative mind map illustrating the differences among AI, ML, DL, Generative AI, and LLMs based on learning approach, data requirements, computational complexity, outputs, and applications. Recommend the best technology for educational chatbots.**

COMPARISON

AI ML DL Generative AI LLM

Rules Learns Neural Creates Understands
Data Networks New Content Language

Automation Prediction Vision Text/Image Chatbot

Educational Chatbot Recommendation: LLM

6. **Design a hierarchical mind map that illustrates the evolution from Artificial Intelligence to Large Language Models. Include important milestones, major algorithms, applications, and dependencies between technologies.**

Artificial Intelligence
|
▼
Machine Learning
|
▼
Deep Learning
|
▼
Generative AI
|
▼
Large Language Models
|
▼
Applications
|
├── **Healthcare**
├── **Education**
├── **Finance**
├── **E-Commerce**
└── **Customer Support**

Major Milestones:

- **AI → Intelligent systems**
- **ML → Learning from data**
- **DL → Deep neural networks**
- **Generative AI → Content creation**
- **LLM → Human-like language understanding and generation**

Rubrics for Calculation

Criteria	Weight age	Level 4 – Exemplary(90–100 Marks)	Level 3 – Proficient(75–89 Marks)	Level 2 – Developing(50–74 Marks)	Level 1 – Beginning(0–49 Marks)
Concept Identification	25%	Identifies all key concepts from literature accurately and comprehensively	Identifies most key concepts with minor omissions	Identifies limited concepts; some are irrelevant or missing	Fails to identify essential concepts

Linkages	25%	Establishes clear, meaningful, and accurate relationships between concepts	Shows correct relationships with minor gaps	Relationships are weak, unclear, or partially incorrect	No meaningful relationships established
Organization	20%	Concepts are well-structured hierarchically with logical flow and grouping	Mostly organized with minor structural issues	Some organization but lacks clear hierarchy	No logical organization or structure
Integration of Literature	20%	Integrates multiple studies effectively to show connections and comparisons	Shows some integration of literature	Limited integration; mostly isolated concepts	No integration of literature evidence
Clarity	10%	Concept map is clear, neat, visually structured, and easy to interpret	Generally clear with minor readability issues	Clarity is reduced; difficult to interpret in parts	Poorly presented and hard to understand